Fiscal Year:	FY 2015	**Task Last Updated:**	FY 06/05/2015
PI Name:	Kozlowski, Steve Ph.D.		
Project Title:	Measuring, Monitoring, and Regulating Teamwork for Long Duration Missions		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **Team**:Risk of Performance and Behavioral Health Decrements Due to Inadequate Cooperation, Coordination, Communication, and Psychosocial Adaptation within a Team		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	I moved from Michigan State University to the University of South Florida in August 2020.		
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Crew Health NNJ12ZSA002N
Start Date:	08/16/2013	**End Date:**	08/15/2016
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	5	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Biswas, Subir (Michigan State University) Chang, Chu-Hsiang (Michigan State University)		
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Performance Goal No.:			
Performance Goal Text:			
	This proposal is for ground-based research: PRD (Program Requirements Document) Risk: Risk of Performance Decrements Due to Inadequate Cooperation, Coordination, Communication, and Psychosocial Adaptation within a Team. IRP (Integrated Research Plan) Gap – Team1: Understand the key threats, indicators, and life cycle of the team for autonomous, long duration, and/or distance exploration missions. Collaboration, cohesion, and coordination are essential teamwork processes, especially for long duration space crews that perform in isolated, confined, and extreme (ICE) environments. Teamwork is critical for minimizing errors and enhancing team performance and reflects team adaptation to the rigors of long duration missions. Over 50 years of research documents the contribution of team processes to team effectiveness. Unfortunately, the vast majority of this research is cross-sectional (static). Thus, there is little scientific knowledge regarding how team processes and psycho-social health vary over long durations in ICE		

Task Description:	conditions, the persistence of disruptive internal and external shocks, and the types of countermeasures that can regulate effective teamwork. The proposed research has three specific aims and deliverables that yield an integrated approach for measuring, monitoring, and regulating teamwork processes and team functioning: (1) Benchmark long duration team functioning in ICE analog environments. This research will use Experience Sampling Methods (daily assessments) to assess team functioning in ICE environments. The goal is to quantify expected variation in key team processes, identify internal and external shocks, and assess dynamic effects on team performance. Such data are essential for developing standards to distinguish normative variation from anomalies indicative of a threat to team functioning which are necessary for triggering countermeasures. (2) Extend engineering development of an unobtrusive monitoring technology (wearable wireless sensor package). The product is to further develop a prototype monitoring technology of teamwork interactions. Initial validation has demonstrated reliability and accuracy sufficient to establish proof of concept. Proposed extensions are designed to (a) add sensing capabilities (swallow monitoring for food intake, stress assessment) and (b) technology development to make the system more robust (packaging, energy efficiency; hardware, algorithms, and software) for out-of-lab field demonstration. (3) Develop teamwork interaction metrics and regulation support systems. The monitoring technology provides continuous data on a range of teamwork processes. Three additional components are required for a countermeasure system. (a) Metrics: Algorithms need to be developed that parse the raw data streams into meaningful measures, then the metrics need to be validated; (b) Data Fusion and Team Regulation Protocols: The multivariate time series metrics need to be fused into a coherent assessment of individual and team functioning. Anomalies that signal a departure from normative functioning have to be classified to drive the provision of feedback and/or team regulation interventions; (c) Distributed Networked Dashboard: A system architecture is needed to integrate sensor information and data fusion, direct feedback to maintain good teamwork and, when the system detects an anomaly in team functioning, trigger appropriate feedback and countermeasures to help an individual or the team regulate team processes. Flexible options for distributing and displaying team status assessments and countermeasures need to be provided (e.g., individual team member, dyads, team leader, ground control). These specific aims will contribute to reducing the risk of team performance decrements by characterizing normative and anomalous patterns of team functioning; monitoring team member interactions; and providing regulation support to maintain teamwork and to trigger countermeasures when needed to aid team recovery.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Team cohesion is not just a critical factor for astronaut teams and ground crews; cohesion is important to the effectiveness of all teams and especially those that operate in critical, high reliability settings. Of the many team process factors that support team effectiveness, team cohesion is the most studied with over a half century of research. Yet, remarkably, very little is known about the characteristics that promote its development and maintenance. For example, we know that experience working together is associated with cohesion formation and maintenance, but what are the mechanisms? Teams that do not cohere replace problematic members or disintegrate so experience only reveals those teams that survive, but that does not tell us why or how. This research, which will uncover the dynamics of collaboration, cohesion, and effective team functioning and will create technologies to monitor team cohesion and guide interventions to restore it, has the potential for wide utility in aviation, military, medical, industrial, and other environments where society depends on the effective performance of high reliability teams.
	At this writing the project has been underway for approximately a year and a half. Our primary efforts have focused on (1) benchmarking data collections in ICE analogs and NASA mission simulations and (2) evaluating the monitoring technology in mission simulations. The other project focus, (3) developing interaction metrics and teamwork support, is dependent on data and techniques developed in (1) and (2). The data from (1) and (2) are just now at a point where we can begin to initiate work on part (3), which will be an area of activity for the next reporting period. However, it is not yet active and so there are no activities to report this year. Benchmark Long Duration Team Functioning in ICE Analog Environments A significant portion of research effort was invested in developing, initiating, and conducting benchmarking data collections over the last several months. A description of our research activities follows. Australian Antarctic Division (AAD) Stations and Field Teams in Aurora Basin (AB). In the prior reporting period, we had extended our collaborative research with Dr. Jeff Ayton of the AAD, gaining approval to continue data collection through the 2016-2017 AAD deployment (assuming that our project continues to be funded by NASA). We initiated new data collections with station personnel who deployed to winter-over for 2014-2015. This involved extending our research protocol with AAD; renewing IRB (Institutional Review Board) approvals by the AAD, Michigan State University (MSU), and NASA; and working with our collaborator and his team to recruit participants. Approximately 46 participants from Mawson, Davis, and Casey Stations, as well as participants from Macquarie Island, are participating in this ongoing effort to benchmark individual and team functioning in ICE settings. This ongoing research assesses daily teamwork processes using Experience Sampling Methodology (ESM), which captures a snapshot of key individual and team reactions to events of the day. Although the absolute sample sizes tend to be small, the primary focus of the research is on the dynamics of reactions over a period of nine months to a year (i.e., approximately 270 to 360 measurement periods), which yields insights into long duration individual and team functioning. To date, we have completed data collection from 26 individuals who wintered over in the Antarctic (Baard, Kermond, Pearce, Ayton, Chang, & Kozlowski, 2014). Participants spent between 9-15 months in their stations and reported a total of 2,333 daily surveys. Descriptive data showed that individuals have different dynamic patterns in terms of their daily reports on different individual and team-level indicators over time. These patterns can be differentiated into four categories: rock solid, uni-varier, multi-varier, or stabilizer. Individuals with a rock solid pattern did not vary in their responses to most of the questions over time. Individuals who are uni-variers primarily varied on one variable, such as daily task or social cohesion. Individuals who are multi-variers varied daily in their responses across a range of

indicators. This may be because they perceived more variability in the environment or because they were more impacted by external factors in the environment. Finally, individuals who are stabilizers initially varied in many variables, but then achieved and maintained a stable equilibrium across the mission.

We also conducted variance decomposition analysis and found that most of the variance in team process indicators (e.g., cohesion, conflict) was at the individual or day level, with little variance attributed to the teams. This indicates that the team level had little meaningful impact and suggests that individuals may not be working closely as a team. The nature of the winter-over crew—individuals with diverse occupations working in isolation from each other—likely explains this pattern. On the other hand, it also means that most of the variability was accounted for by the person and environment.

Random coefficients modeling revealed that relationships between some variables were reciprocal, while relationships between other variables were directional. Cohesion and performance had reciprocal, positive relationship with each other. This suggests that prior cohesion predicted future performance positively, and prior performance positively contributed to future cohesion ratings. Positive affect and team conflict had reciprocal, negative relationship with each other, suggesting that prior positive mood was related to less future conflict, and conflict was predictive of lower positive mood. On the other hand, conflict management had directional relationship with cohesion, such that prior conflict management was positively related to subsequent team cohesion. This finding suggests that teams engaging in conflict management felt more cohesive the next day, but that being cohesive may not necessarily make the team more effective at managing their conflict. Finally, prior positive affect also had directional, negative relationship with subsequent negative affect, supporting the buffering role of positive mood.

Although (given the sample size) we consider these finding preliminary, they suggest that selection of individuals for winter-over deployments should consider the utility of assessing for positive affectivity, an individual disposition (i.e., personality characteristic) that supports positive mood states. It also suggests that conflict management skills, bolstered by training, could help individuals and teams maintain effective working relationships across the long span of winter-over deployments to the Antarctic.

Science Field Teams in Antarctica. We also extended our ongoing collaboration with science teams that deploy to the ice during the Antarctic summer for 2014-2015. This marks our fifth season of data collection with this research team. This involved extending our research protocol, renewing MSU and NASA IRB approvals, and recruiting participants from the science teams. Approximately 7 participants are contributing to the research effort, providing daily ESM reports.

Results based on the data collected to date suggest that teams vary in the key indicators of team processes both in terms of the levels (e.g., average cohesion across members) and the degree of agreement (e.g., standard error of cohesion across members; Pearce, Baard, Harvey, Karner, Chang, & Kozlowski, 2014; Pearce, Baard-Perry, Harvey, Karner, & Ayton, 2015). Moreover, cohesion and adaptability showed reciprocal relationships over time, such that higher cohesion on the prior day predicted higher adaptability on the subsequent day and vice versa. In addition, cohesion and negative affect also showed reciprocal relationships over time, such that high negative affect on the prior day predicted low cohesion the subsequent day and vice versa. These results suggest that high cohesion may be crucial in buffering the negative effects of poor individual psychosocial well-being on team effectiveness.

Human Exploration Research Analog (HERA). We continued benchmarking research in HERA, a NASA mission simulation located at the Johnson Space Center (JSC), that was initiated in 2014. HERA missions involve a crew of 4 members, selected from NASA volunteers. HERA simulates a transit mission for exploration of an asteroid. In Campaign 1, mission duration was approximately 7 days for 4 crews of 4 members each. Campaign 2, initiated in January 2015, extended the missions to 14 days. This research has involved extending our protocol, securing IRB approvals from NASA and MSU, training personnel, and coordinating research activities with several other investigator teams. We also have taken on the responsibility of coordinating several end-of-day measures across investigators and then compiling and sharing the data. We are also the lead team for coordinating interaction badge data (the “SS” badge provides shared data; the other MSU badge is under development and evaluation).

Task Progress:

In addition to the use of our standard ESM protocol, we also employ a “simulation within the simulation” that is used to evaluate our monitoring technology. Heretofore, the monitoring “badge” has only been evaluated in lab settings for basic validation. This effort is extending evaluation for field testing and user reactions. Since the last report, we have collected data from crew members of two HERA missions in Campaign 1 in June and September of 2014. We have also collected data for two HERA missions in Campaign 2 in January/February and April of 2015. Data collection for the third HERA mission of Campaign 2 will begin in June of 2015.

We found that based on the data collected from HERA Campaign 1 (106 daily reports from 16 participants), different teams had different experiences throughout the mission and responded differently to the same environmental stressors (e.g., sleep deprivation, communication delays; Dixon & Vessey, 2015). In addition, analysis of agreement between crew members’ ratings on team processes indicated that within the same team, individual members do not always view their team or their shared experiences the same way. Finally, across the four missions in Campaign 1, there was no clear pattern of reactions to experiences across teams that can be identified, despite the seemingly similar exposure to the simulated stimuli. These preliminary results show that the continued data collections with more teams in the HERA environment will be informative of how different teams and individuals experience these environments.

Hawai‘i Space Exploration Analog and Simulation (HI-SEAS). We continued benchmarking research in a surface exploration simulation, HI-SEAS, which is located at 8200 feet on Mt. Mona Loa on the big island of Hawai‘i, and was initiated in the prior reporting period. This has involved extending our protocol; securing IRB approvals from the University of Hawai’i (under the PI, Kim Binsted), MSU, and NASA; and substantially aiding the HI-SEAS mission design. We have contributed to crew selection (we screened on the five factor model of personality and cognitive ability), the mission story / script, and mission EVA (extravehicular activity) / scenario design.

We completed the data collection from the five-person crew of mission 1, and are currently collecting ESM data from the six-person crew during their 240-day mission that began in October 2014. The crew is also using the SS badge and the MSU monitoring badge so that we can enlarge the pool of benchmarking data for interactions over time. We are now preparing for the next HI-SEAS mission that will involve a crew of 6 for 360 days. The mission is set to begin in the fall of 2015.

Overall, the five-person crew from mission 1 provided a total of 485 daily surveys (Santoro & Binsted, 2015). Random coefficient modeling showed that team processes were reciprocally related to one another. Positive autoregressive effects

	were found for the indicators for individual members' psychosocial health (e.g., positive and negative affect) and other team process and effectiveness indicators (e.g., cohesion, conflict, performance) from the previous day to the next. In addition, cohesion played a major role in affecting other team processes, such as positively impacting performance and affect. Finally, improvements in certain team processes and individual psychosocial well-being (i.e., cohesion, negative affect, and positive affect) from one day to the next benefitted team effectiveness on the next day. Thus, similar to the long duration findings for the AAD stations, we see evidence that positive affect (which can be selected based on individual dispositions) and cohesion (which can be bolstered by team interactions and team leadership) have buffering effects on negative factors that impede individual and team psychosocial health. NASA Extreme Environment Mission Operations (NEEMO) Mission 18. Finally, we conducted data collection from a crew of four astronauts and two habitat technicians for NEEMO18. The roughly one week mission was completed in July 2014. The astronauts represented a mix of international space agencies, so the crew was multicultural. NEEMO18 data collection involved extension / modification of our protocol; coordinating IRB approvals with NASA, MSU, and several international space agencies; training revisions; and coordination among investigating teams and NASA elements. We were, once again, responsible for coordinating end-of-day reactions and badge data. The NEEMO data have largely been helpful for feedback on user reactions to the badges. Extend Engineering Development of an Unobtrusive Monitoring Technology The monitoring technology under development has been successfully validated in the laboratory and is now under evaluation in NASA mission simulations. Primary objectives for engineering development center on: (1) improved packaging and enhancing robustness of the wearable interaction monitoring badges and (2) detection of swallow monitoring using the wearable sensor system. Packaging and robustness. We have developed a 3-D printed case for the wearable badge. The case contributes to robustness in terms of the badge abilities to withstand rough handling in variable environmental conditions. The access point has also been repackaged. The new access point is also capable of all the sensing possible using the badges. To improve energy-efficiency, we have employed a series of transmission power control protocols (developed by Co-PI Biswas' group) running on the badge hardware. A novel measurement based link power control mechanism with closed-loop feedback control techniques was used. These protocols have improved the battery life of the system. The current version of the badge with all these new software and protocol can run up to 6 hours of data collection in one recharge. These new robust badges were used in the HERA data collection sessions during this reporting period. Swallow monitoring. We have developed a wearable solid food intake monitoring system that analyzes human breathing signal and swallow sequence locality for solid food intake monitoring. Food intake is identified by the way of detecting a person's swallow events. The system works based on a key observation that the otherwise continuous breathing process is interrupted by a short apnea during swallowing. A Support Vector Machine (SVM) is first used for detecting such apneas in breathing signals collected from a wearable chest-belt. The resulting swallow detection is then refined using a Hidden Markov Model (HMM) based mechanism that leverages known locality in the sequence of human swallows. Using the developed system in this reporting period we are experimentally able to demonstrate the effectiveness of such two-stage SVM-HMM based mechanism for solid food intake detection via analyzing breathing signal and human swallow sequence locality. Apnea detection also has potential as an additional data modality for assessing stress during team member interactions. As this badge capability develops, it will be integrated into our phased lab validation process. Flexible badge identifier. The embedded software on the badges and on the access point has been upgraded to support dynamic determination of badge identification. In the earlier version, each badge had to be hardwired with a unique badge identifier. In contrast to the earlier version, the newly developed software allows each badge to be allocated an identifier by the access point at start time (when the system is booted). This feature brings a huge operational advantage in that any badge can be used with any identifier by turning it on in a given sequence. Currently, the engineering team is working to upgrade the software for the Graphical User Interface so that the flexible identification feature can be employed by end-users.
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